

# SANYAM BOHRA

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## TECHNICAL SKILLS

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- **Languages:** C++ (DSA), Python, JavaScript (ES6+), HTML5, CSS3, SQL
- **AI & Machine Learning:** TensorFlow, Scikit-Learn, OpenCV, NumPy, Pandas, Matplotlib, Seaborn, TLE Data Processing
- **Web Development:** React.js, Node.js, Express.js, Firebase
- **Developer Tools:** Git, GitHub, VS Code, Linux, LaTeX
- **Core Fundamentals:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP)

## EDUCATION

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|-----------------------------------------------------------------------|---------------------|
| <b>MBM University, Jodhpur</b>                                        | Aug 2024 – May 2028 |
| Bachelor of Engineering (B.E.) in AI & Data Science                   | <b>CGPA: 7.22</b>   |
| <b>Senior Secondary Education (Class XII)</b>                         | 2023                |
| Percentage: 86%   JEE Mains: 95.64 Percentile (National Rank: Top 5%) |                     |

## LEADERSHIP & EXPERIENCE

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| <b>Embedded Systems and Robotics Club (ESRC), MBM University</b> | Oct 2024 – Present |
| <i>Core Member</i>                                               |                    |

- Collaborating on technical projects involving hardware-software integration and robotic automation.
- Organizing university-level workshops on microcontrollers and embedded programming for students.
- Developing and optimizing C++ scripts for sensor data acquisition and real-time robotic response.

## TECHNICAL PROJECTS

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**Space Debris Detection & Trajectory Tracking** | *Python, Scikit-Learn* [GitHub Link](#)

- Engineered a machine learning system to classify orbital objects using Two-Line Element (TLE) data.
- Extracted features from orbital parameters like inclination and eccentricity to distinguish debris
- Implemented a Random Forest Classifier to predict collision risks in Low Earth Orbit (LEO).

**Automated Space Object Detection** | *Python, OpenCV, TensorFlow* [GitHub Link](#)

- Developed a computer vision pipeline using OpenCV for image preprocessing and noise reduction. 26]

- Built and trained a CNN for identification of man-made objects in satellite imagery.
- Utilized Seaborn for data visualization of debris density across different orbital altitudes.

#### **Career Guidance Prototype** | *React.js, Node.js, Firebase*

- Developed a full-stack platform featuring personalized career roadmaps and user skill assessments.
- Integrated Firebase Authentication and NoSQL database for secure, real-time data management.
- Managed application state using React Hooks to provide a responsive and intuitive user interface.